SMART EXPENSE SPLITTER AND TRACKER

Track Smarter. Split Fairer. Save Better.

Group 28

PROJECT OVERVIEW

A smart expense management platform that brings personal expense tracking, group bill splitting, budgeting, and AI-powered financial insights together in one place.

KEY STAKEHOLDERS

PRIMARY

Individuals
Students and working professionals managing personal expenses.

Groups
Friends, roommates, and colleagues splitting shared bills.

Group administrators
Managing group members and shared expenses.

SECONDARY

- Project owners
- Development team
- Customer support team

MAIN FEATURES

01 Personal Expense Tracking
Add, edit, categorise, and analyse expenses.

02 Group Expense Splitting
Divide bills fairly and track outstanding balances.

03 Budget Management
Set spending limits and receive budget alerts.

04 AI Financial Advisor
Get spending insights and personalised saving suggestions.

05 Reports and Analytics
Understand spending patterns through charts and summaries.

SYSTEM WORKFLOW

1. Register / Login
2. Add Personal or Group Expense
3. Categorise and Split Expenses
4. Track Budgets and Balances
5. View Reports and AI Insights

SOFTWARE REQUIREMENTS

Functional

- User registration and login
- Personal and shared expense tracking
- Expense categorisation and analytics
- Group splitting and settlements
- AI-powered financial advice
- Budget alerts and recurring expenses
- Notifications

Non-Functional

- **Usability:** Easy to learn and navigate.
- **Performance:** Responsive interactions.
- **Security:** Protect users' financial information.
- **Reliability:** Handle errors without losing user data.
- **Maintainability:** Keep the code organised and easy to update.
- **Scalability:** Support growth in users and data.

IMPORTANT DESIGN PRECAUTIONS

- AI suggestions must never override a category manually selected by the user.
- Recurring group expenses must have an explicit automatic-splitting option.
- AI features must protect financial data through secure processing and data minimisation.

AGILE SPRINT PLAN *Planned development activities*

1 — Sprint 1
Requirements, Design and AI Feasibility

2 — Sprint 2
Authentication and Backend Setup

3 — Sprint 3
Personal Expense Tracking and Analytics

4 — Sprint 4
Group Expense Splitting

5 — Sprint 5
AI Integration and PWA Features

6 — Sprint 6
Integration, Testing and Deployment

EXPECTED OUTCOMES

- ✓ Easier personal expense management.
- ✓ Fair and transparent group settlements.
- ✓ Better budgeting and savings awareness.
- ✓ Useful, personalised financial insights.